

SPECIFICATION

Product Name: Integrated Air Quality Sensor Module

Item No.: AM2009DY/AM2020DY

Version: V0.5

Date: Feb 7th, 2026

Revision

No. Version	Content	Date
V0.1	The first version	2025-09-07
V0.2	Add communication protocol response content	2025-10-17
V0.3	Add fault detection code in communication protocol Delete VOC/NO2/HCHO output mask description in communication protocol	2025-11-28
V0.4	Update communication protocol Update specifications content	2025-12-16
V0.5	1.Update datasheet measurement range and 2D dimensions 2.Update the recommended and non-recommended installation 3.Add datasheet parameters 4.Add user attention content	2025-02-07

Integrated Air Quality Sensor Module

AM2009DY/AM2020DY



Applications

- Air purifier
- Air quality monitor
- Air conditioner
- Ventilation system
- Consumer electronic products
- Environmental monitoring

Description

AM2009DY is an integrated air quality sensor module with function of particle concentration, VOC and NO₂ concentration, temperature and relative humidity measurement. AM2020DY is with extra HCHO module. Particle concentration is measured by laser particle sensor which uses light scattering principle. It measures and calculates the suspended particle number exactly, and output particle mass concentration $\mu\text{g}/\text{m}^3$ directly via mathematical algorithm and scientific calibration. VOC&NO₂ concentration are measured by a combined MOX sensor, which can detect VOC and NO₂ simultaneously. It has extremely high sensitivity to various kinds of organic volatile gases such as benzene, ethanol, formic acid, nicotine and some inorganic gases like carbon monoxide, ammonia and hydrogen. HCHO sensor is an electrochemical gas concentration sensor.

Features

- Super highly integrated, with function of particle, VOC, NO₂, temperature and humidity measurement
- The smallest particle size of available measurement: 0.3 μm
- Real-time output particle mass concentration in $\mu\text{g}/\text{m}^3$ available
- High accuracy, high sensitive and quick response
- Temperature compensation algorithm
- Matrix calibration technology and optimized algorithm, for high accuracy and consistency
- Siloxane resistance of VOC sensor, for long lifespan and high stability
- Intelligent VOC auto-calibration strategy on site, maintenance-free

Working Principle

- The PM sensor integrated adopts laser scattering technology to detect particle mass concentration
- The VOC&NO₂ sensor is based on MOX technology, HCHO sensor is based on electrochemical technology.
- The RH&T sensor integrated adopts capacitance resistance materials to detect indoor RH&T

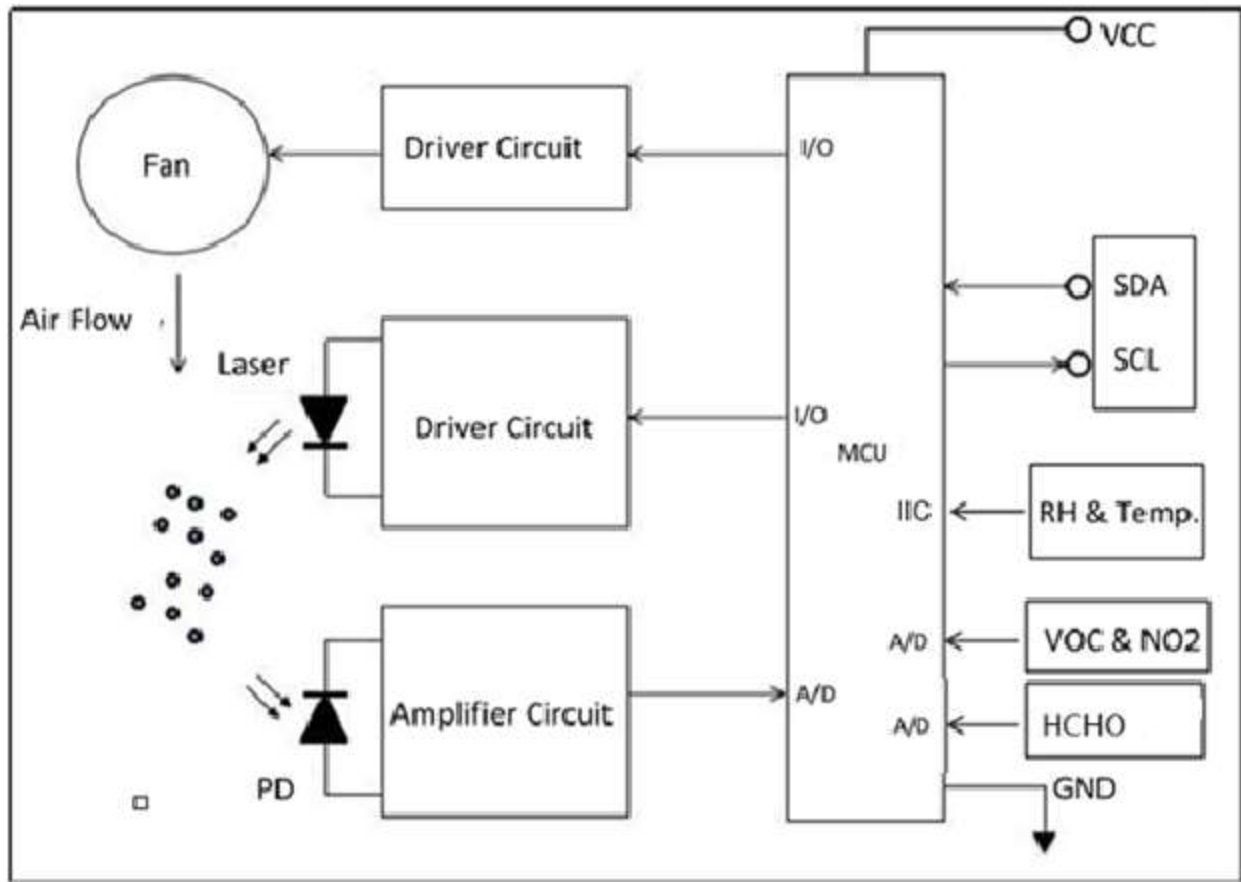
Specifications

Integrated Indoor Air Quality Sensor Specification	
Operating principle	Laser scattering/MOX/Electrochemical
Measurement range	Particle: 0~1,000 $\mu\text{g}/\text{m}^3$; VOC: 0~10ppm, NO ₂ : 0~1,000ppb, HCHO: 0~1000ppb Temperature: -20~70°C, Humidity: 0~99%RH
Measured particle range	0.3 μm ~10 μm
Output resolution	Particle: 1 $\mu\text{g}/\text{m}^3$, VOC: 1ppb, NO ₂ : 1ppb, HCHO: 1ppb Temperature: 0.1°C, Humidity: 0.1%RH
Working condition	-10°C ~ 50°C, 10~95%RH (non-condensing)
Storage condition	-20°C ~ 60°C, 10~95%RH (non-condensing)
Particle accuracy	PM1.0/PM2.5: 0~100 $\mu\text{g}/\text{m}^3$: $\pm 10\mu\text{g}/\text{m}^3$; 101~1000 $\mu\text{g}/\text{m}^3$: $\pm 10\%$ reading PM10: 0~100 $\mu\text{g}/\text{m}^3$: $\pm 20\mu\text{g}/\text{m}^3$; 101~1000 $\mu\text{g}/\text{m}^3$: $\pm 20\%$ reading (Reference: TSI8533, Dust Source: DEHS+Arizona A1 Dust, @condition 25 $\pm 2^\circ\text{C}$ & 50 $\pm 10\%$ RH)
Temperature accuracy ¹	$\pm 1^\circ\text{C}$ (0~50°C)
Humidity accuracy ¹	Typical/Max: $\pm 5\%/\pm 8\%$ (5%~95%RH) ²
VOC Accuracy	200ppb or $\pm 50\%$ of reading, whichever is larger (25 $\pm 2^\circ\text{C}$, 50 $\pm 10\%$ RH)
NO ₂ Accuracy	50ppb or $\pm 50\%$ of reading, whichever is larger (25 $\pm 2^\circ\text{C}$, 50 $\pm 10\%$ RH)
HCHO Accuracy(AM2020)	$\pm 25\text{ppb}$ or $\pm 25\%$ of reading, whichever is larger (Reference instrument: Gasera one, 25 $\pm 2^\circ\text{C}$, 50 $\pm 10\%$ RH)
Cross sensitivity to ethanol(HCHO in AM2020)	<0.5%
Sampling interval	1s (PM, VOC, NO ₂ , Temperature & Humidity)
Startup time	≤ 1800 seconds (VOC&NO ₂); ≤ 8 seconds (Particle); ≤ 8 seconds (RH&T) T90<120s(HCHO)
Power supply	DC 3.3V $\pm 0.1\text{V}$ ~5V $\pm 0.1\text{V}$, ripple wave < 100mV
Average working current	$\leq 180\text{mA}$ (continuous operation)
Laser wavelength	940nm
ESD Rating	Electrostatic Discharge: $\pm 4\text{kV}$ contact, $\pm 4\text{kV}$ air
Acoustic Emission Level	<25dB(A) @30cm; Drift value is <3dB(A) in lifetime
Dimensions	55.3*19.95*26mm
Weight	<25g (To be finalised)
Major Firmware Version	V2.20.1.9X
Digital output	IIC
Life time	≥ 10 years (24h/day operation)

Note:

1. Temperature and humidity accuracy in the table are defined at stable environment;
2. Humidity accuracy is defined at temperature of 25 $\pm 2^\circ\text{C}$;

Internal Architecture Description



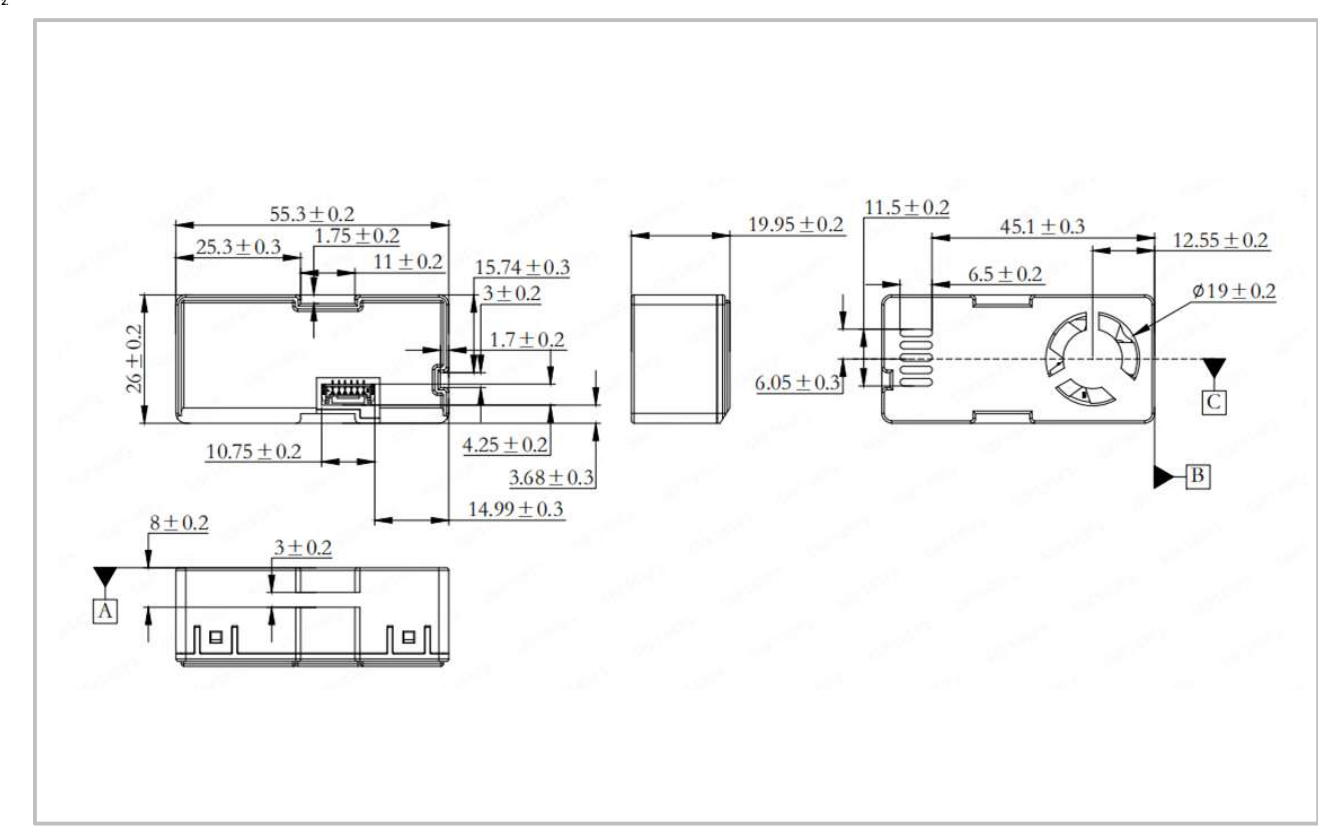
According to the above figure, the light source part of AM2009DY/AM2020DY is composed of a laser tube and a driving circuit. The detection part of the sensor is composed of light sensitive part which receives reflected light and amplifying circuit. Data processing and communication output are completed by microprocessor.

The gas flows into the module through the fan, when sampling particles pass through light beam (laser), there will be light scattering phenomenon, and scattered light will be converted into electrical signal (pulse) via light sensitive part. Electrical signal will be transformed into digital signals after amplifying circuit, smoothing and MCU processed.

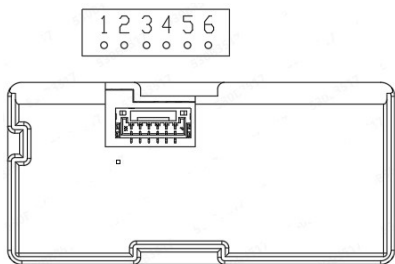
Temperature and relative humidity is measured by sensor directly and data is transferred to MCU, and it is with compensation. MOX type VOC&NO2 sensor is integrated, which has extremely high sensitivity to NO2 gas and various kinds of organic volatile gases such as formaldehyde, benzene, ethanol, formic acid, nicotine and some inorganic gases like carbon monoxide, ammonia and hydrogen.

Dimensions and Connector

1. Dimensions (Unit mm, tolerance ± 0.2 mm)



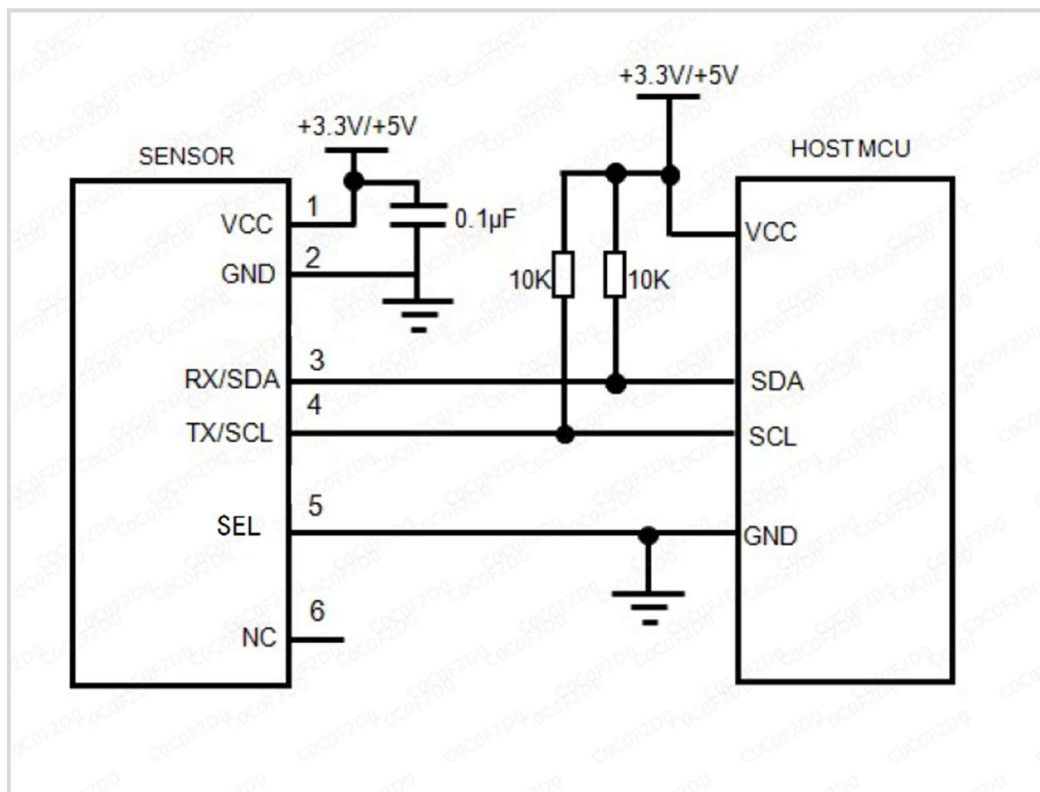
3. I/O Connector Pin out



No.	Pin	Description
1	Supply Voltage	3.3V
2	GND	Ground
3	Serial data input/output	SDA for IIC communication
4	Serial clock input	SCL for IIC communication
5	Interface selection	Output-mode exchange: @3.3V High-level or floating is UART communication mode; low level is IIC communication mode
6	NC	Do not connect (For manufacture internal use only)

Item	Part Number	Pitch	Available Mating Connector
Connector	ACES 51468-0064N-001	1.25 mm	ACES 51452-006H0H0-001)

Typical Application Circuit



Note of Circuit Design

- IIC communication level depends on customer pull-up level (3.3V~5V compatible).
- PIN6 should be floating for normal working.
- Power supply of sensor is suggested to be 3.3V, please refer to specification for more details.
- For other pull-up resistor options, please add the pull-up resistor on the HOST main PCB.

Communication Protocol

1. General Statement

- This is IIC protocol for AM2009DY/AM2020DY. The sensor module is lower computer, which is not able to initiate communication automatically. Communication is initiated via main control board, which reads data and sends control commands;
- Communication clock frequency $\leq 100\text{KHz}$;
- Original IIC address of AM2009DY/AM2020DY is 0x28(7-bit), when add writing bit, it's 0x50, and when add reading bit it's 0x51.

2. Format of Communication Protocol

START: start signal, send by main control board;

STOP: stop signal, send by main control board;

ACK: acknowledge signal, send by the sensor module if in bold; otherwise, send by main control board;

NCK: non-acknowledge signal, send by the sensor module if in bold; otherwise, send by main control board;

Px: receive and send data; send by the sensor module if in bold; otherwise, send by main control board.

3. Protocol Detailed Description

3.1 Send Control Command

Send by main control board:

START+WRITE+**ACK**+P1+**ACK**+P2+**ACK**..... +Pn+**ACK**+CS+**ACK**+STOP

Byte	Content	Description
Address	Sensor address + reading / writing bit	0x50 for writing
P1	0x11	Head
P2	Frame length	Frame length=CMD+total DATA length
P3	command	command
P4	Data	Data
...	...	Data
...	...	Data
CS	Checksum	Checksum=256 - (head +frame length + command +data)

3.2 Send Reading Command

Send by main control board:

START+READ+**ACK**+P1+**ACK**+P2+**ACK**.....

Byte	Content	Description
Address	Sensor address + reading / writing bit	0x51 for reading
P1	0x16	Head
P2	Frame length	Frame length=CMD(length is 1) + total DATA length
P3	Command	Command
P4	Data	Data
...	...	Data

CS	Checksum	Checksum=256 - (head +frame length + command +data)
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3.3 Command table

Item No.	Function Description	Command
1	Read measurement result	0x16
2	Read software version	0x1E
3	Read serial number	0x1F
4	Read sensor model	0x2E
5	Open/Close fan and laser diode	0x0C
6	Manual zero calibration of VOC and NO2	0x360B

4. Detail Description of Protocol

2.1 Read Measurement Result

Send: 11 01 16 D8

Response: 16 17 16 DF1~DF22 [CS] (for AM2009/AM2020)

Response description :

- DF1-DF2: TVOC concentration = $DF1 \times 256^1 + DF2$, data shall transfer from hex. to dec., unit: ppb
- DF3-DF4: NO2 concentration = $DF3 \times 256^1 + DF4$, data shall transfer from hex. to dec., unit: ppb
- DF5-DF6: PM1.0 = $DF5 \times 256^1 + DF6$, data shall transfer from hex. to dec., unit: $\mu\text{g}/\text{m}^3$
- DF7-DF8: PM2.5 = $DF7 \times 256^1 + DF8$, data shall transfer from hex. to dec., unit: $\mu\text{g}/\text{m}^3$
- DF9-DF10: PM10 = $DF9 \times 256^1 + DF10$, data shall transfer from hex. to dec., unit: $\mu\text{g}/\text{m}^3$
- DF11-DF12: Temperature = $DF11 \times 256^1 + DF12$, data shall transfer from hex. to dec., unit: $^{\circ}\text{C}$
(compensated real temperature $((DF11 \times 256^1 + DF12) - 500) / 10$)
- DF13-DF14: humidity = $DF13 \times 256^1 + DF14$, data shall transfer from hex. to dec., unit: 1% (compensated real relative humidity $((DF13 \times 256^1 + DF14) / 10)$)
- DF15-DF16: TVOC ADC = $(DF15 \times 256^1 + DF16) / 10$, data shall transfer from hex. to dec.
- DF17-DF18: NO2 ADC = $(DF17 \times 256^1 + DF18) / 10$, data shall transfer from hex. to dec.
- DF19-DF20: HCHO concentration = $DF19 \times 256^1 + DF20$, data shall transfer from hex. to dec. unit: ppb
(HCHO output reading is only for AM2020DY)

11. DF21-DF22: Fault detection code

Note: 01 is for abnormal fault, and 00 is for normal performance case.

DF21~DF22:

Bit	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Fault Detect Code	Fan speed Faulty: RPM is out of 2250~9000 range	PM signal Faulty: Photodiode signal > 4000AD	VOC working temperature Faulty: VOC working temp is out of 320~450 $^{\circ}\text{C}$ range	VOC sensitive material Faulty: VOC sensitive resistance is out of 1k~40.94M Ω range	NO2 working temperature Faulty: NO2 working temp is out of 200~350 $^{\circ}\text{C}$ range	NO2 sensitive material Faulty: NO2 sensitive resistance is out of 1k~40.94M Ω range	HCHO signal channel Faulty: Signal > 40000	HCHO reference channel Faulty: Signal is out of 15000~21000 range
Bit	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Fault Detect Code	RHT Faulty: IIC communication is failed or CRC is failed	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

Faulty detection code explanations:

- Faulty alarm code triggering strategy: Corresponding parameter is out of threshold value continuously longer than 10s
- Faulty alarm code cleaning strategy: Detecting the corresponding parameter is with the normal threshold range is longer than 1s

Example on Cubic Testing Board:

Send: 50 11 01 16 D8 00 1A // Note: the last two bytes are for Cubic testing board, 00 is starting symbol, and 1A is byte digit need to be read, which means response will be 26 bytes, for other device setting method might be different.

Send: 51 // Note: Typical interval 50ms, which means 51 shall be sent 50ms later since previous command.

Response: 16 17 16 01 47 00 40 00 45 00 4A 00 58 02 8B 01 95 40 92 7E FA 00 3C 00 00 CS // Note: Response time = 65ms on Cubic testing board

VOC concentration= $1 \times 256 + \text{hex2dec}(47) = 327\text{ppb}$;

NO2 concentration= $0 \times 256 + \text{hex2dec}(40) = 64\text{ppb}$;

Particle concentration: PM1.0= $\text{hex2dec}(45) = 65\mu\text{g}/\text{m}^3$, PM2.5= $\text{hex2dec}(4A) = 70\mu\text{g}/\text{m}^3$, PM10= $\text{hex2dec}(58) = 88\mu\text{g}/\text{m}^3$;

Temperature= $[2 \times 256 + \text{hex2dec}(8B) - 500] / 10 = 15.1^\circ\text{C}$; Humidity= $[1 \times 256 + \text{hex2dec}(95)] / 10 = 40.5\%$.

VOC ADC= $64 \times 256 + \text{hex2dec}(92) = 16530$; NO2 ADC= $126 \times 256 + \text{hex2dec}(FA) = 32506$

HCHO concentration= $00 \times 256 + \text{hex2dec}(3C) = 60\text{ppb}$

Note: hex2dec in above calculation means transfer hexadecimal value to decimal value.

2.2 Read Software Version

Send: 11 01 1E D0

Response: 16 0E 1E DF1~DF13 [CS]

Function: Read software version

Note:

Software version="DF1~DF13"

Should change the HEX code to ASCII code.

Example:

HEX code: 16 0E 1E 50 4D 20 56 31 2E 31 39 2E 30 2E 30 33 F3

Transfer DF1~DF13 to ASCII code: PM V1.19.0.03

Example on Cubic Testing Board:

Send: 50 11 01 1E D0 00 11 // Note: the last two bytes are for Cubic testing board, 00 is starting symbol, and 11 is byte digit to be read, which means the response will be 17 bytes.

Send: 51 // Note: Typical interval 50ms, which means 51 shall be sent 50ms later since previous command.

Response: 16 0E 1E 50 4D 20 56 31 2E 31 39 2E 30 2E 30 33 F3 // Note: Response time = 70ms.

HEX code: 50 4D 20 56 31 2E 31 39 2E 30 2E 30 33, transfer to ASCII code is: PM V1.19.0.03

2.3 Read Serial Number

Send: 11 01 1F CF

Response: 16 0B 1F DF1 DF2 DF3 DF4 DF5 DF6 DF7 DF8 DF9 DF10 CS

Function: Read serial number

Note:

Serial number = (DF1*256+DF2), (DF3*256+DF4), (DF5*256+DF6), (DF7*256+DF8), (DF9*256+DF10)

Example:

Response: 16 0B 1F 00 00 00 7E 09 07 07 0E 0D 72 9E

Serial number: 0126 2311 1806 3442

Example on Cubic Testing Board:

Send: 50 11 01 1F CF 00 0E // Note: the last two bytes are for Cubic testing board, 00 is starting byte, and 0E means the response is 14 bytes.

Send: 51 // Note: Typical interval 50ms, which means 51 shall be sent 50ms later since previous command.

Response: 16 0B 1F 00 00 00 37 06 54 13 C2 00 3C 1E // Response time is 60ms

Transfer DF1~DF10 to serial number: 0000 0055 1620 5058 0060

2.4 Read Sensor Model

Send: 11 01 2E C0

Response: 16 07 2E DF1 DF2 DF3 DF4 DF5 DF6 DF7 DF8 CS

Function: Read sensor model, DF1~DF8 is ASCII code

Note:

Sensor Model="DF1~DF8"

Should change the HEX code to ASCII code.

Example on Cubic Testing Board:

Send: 50 11 01 2E C0 00 0C // Note: the last two bytes are for Cubic testing board, 00 is starting symbol, and 0C means the response is 12 bytes

Send: 51 // Note: Typical interval 50ms, which means 51 shall be sent 50ms later since previous command.

Response: 16 09 2E 41 4D 32 30 30 39 44 59 CS//Note: Response time =

65ms Sensor Model:

Transfer to HEX is 41 4D 32 30 30 39 44 59, correspond ASCII code is AM2009DY

2.5 Turn off/on Fan and Laser Diode

Send: 11 03 0C DF1 1E CS

Function: Turn off/on fan and laser diode

Note:

1. DF1=02 means turning on fan and laser diode, DF1=01 means turning off fan and laser diode;
2. When re-power on the sensor, fan and laser diode will be turned on by default.

Example on Cubic Testing Board:

Send: 50 11 03 0C 01 1E C1 00 06 //Turn off fan and laser diode, typical response time = 50ms

Send: 50 11 03 0C 02 1E C0 00 06 // Turn on fan and laser diode, typical response time = 50ms

Note: in the above commands, 00 is starting symbol of Cubic testing board, and 06 is byte digit to be written.

2.6 Manual Zero Calibration of VOC and NO2

Send: 11 06 36 0B 00 DF1 00 00 CS

Function: Zero calibration of VOC and NO2

DF1=0xF4 for VOC zeroing, DF1=0xFE for NO2 zeroing

Note:

1. Manual zero calibration of VOC and NO2 shall be performed in fresh air, in which no VOC or NO2 exist;

Example on Cubic Testing Board:

Send: 50 11 06 36 0B 00 FE 00 00 AA 00 09 //NO2 zeroing

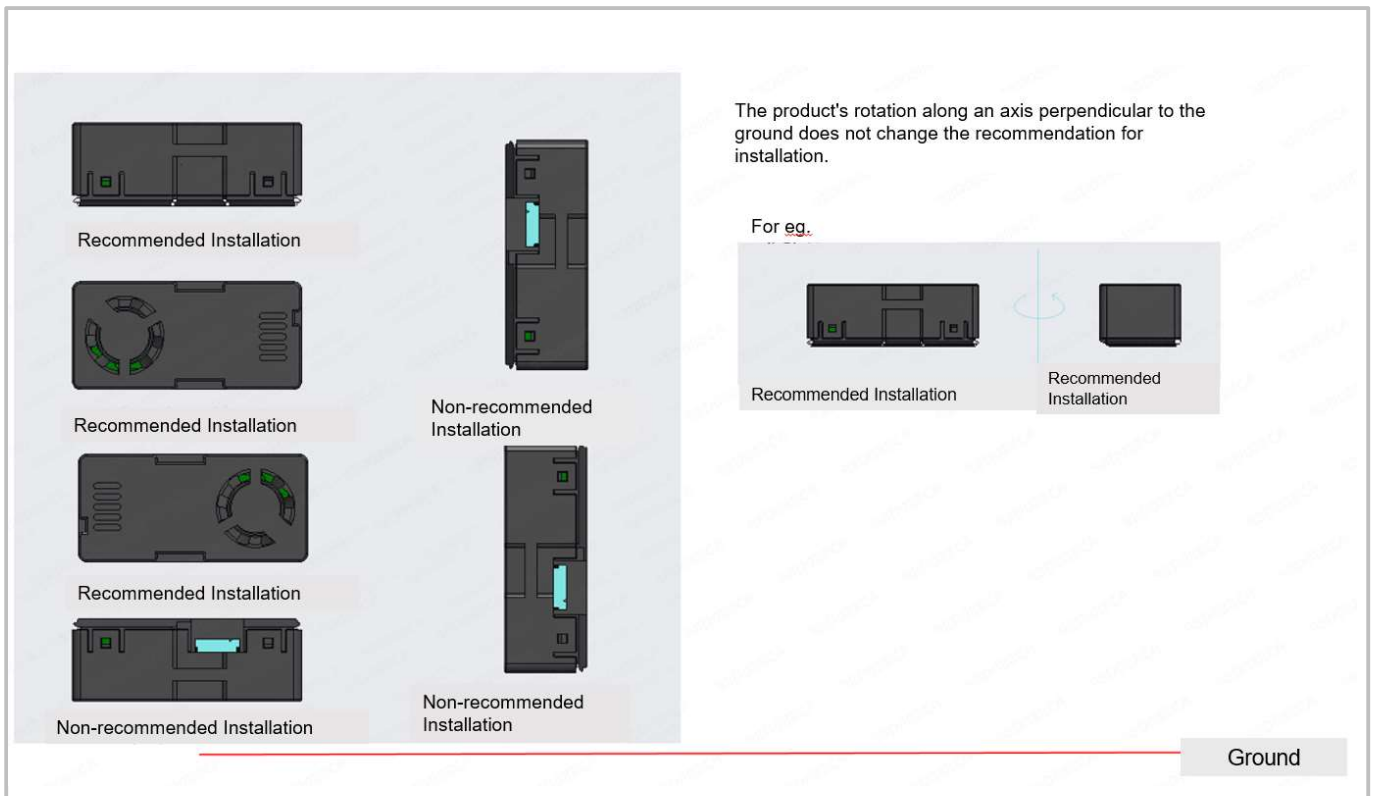
Send: 50 11 06 36 0B 00 F4 00 00 B4 00 09 //VOC zeroing

Note: in the above commands, 00 is starting symbol of Cubic testing board, and 09 is byte digit to be written

User Attention

- The sensor is for household electronics products. For application of medical, mining, disaster preparedness, which needs high security and high dependence, this sensor is not suitable.
- Please do not use it in bad dusty environment and close sampling port.
- Avoid using the sensor under situation with strong magnetic, such as situation close to stereo speaker, microwave oven, induction cooking. Please observe electrostatic discharge (ESD) protection precautions when assembling the sensor;
- Avoid exposure to volatile silicon compound vapors and high concentrations of organic solvents to prevent poisoning of the sensor's sensitive material;
- Avoid prolonged exposure to high temperature and high humidity environments during power-on operation to prevent rapid degradation of gas sensitivity;
- After long-term power-off, allow the sensor to power on for a period of time before testing; the longer the power-off duration, the longer the required power-on time;
- The sensor's sensitivity is affected by temperature and humidity; when integrated into a module, it can be paired with a temperature and humidity sensor for algorithm-based compensation.
- When install the sensor, make sure the inlet and outlet is unobstructed, and cannot be fronted onto large air stream. There are two sides cannot be put downwards (As below pictures), In case of dust deposition on the surface of sensitive device, dust deposition will affect accuracy of sensor.

Recommended installation and Non-Recommended installation



- The case of sensor connects with the DC ground of inner circuit directly, which will cause safety problem if touching with DC ground. To avoid this problem, Sensor should be internally installed and no permit for touching sensor before power off.
- There is no high-pressure transient protection circuit of the sensor. The power supply of the sensor should be stable

5V/3.3V and low noise. Please refer to the working current in specification table.

- The sensor itself is safe to use. What the user should be cautious is the safety of power supply and structure design on the sensor.
- This laser diode inside is defined as class 3R laser product according to GB7247.1-2012 or IEC60825-1:2014 standard. Please do not disassemble the sensor, to avoid laser light direct illumination to the eye.

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